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A U S T R A L I A

INFS7450 SOCIAL MEDIA ANALYTICS (ST LUCIA) - SEMESTER 1, 2026

Project One

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Abstract

The present document reports the results of the first project of the course INFS7450 Social Media Analytics [1]. The objective of the project was to load, visualize, and analyze an extraction of a *Facebook* network through two key techniques presented in the course. Indeed, despite the existence of numerous libraries (in this case, *NetworkX* [2] was used as ground truth), manually writing these algorithms remains the only way to fully comprehend the underlying concepts that power everyday graph algorithms. In a broader sense, the present study aims to demonstrate a practical illustration of a graph analysis in a real-world context. This objective was pursued by employing authentic data and extensive handwritten documentation. The provided code [3] first loads the network and presents an overview of the nodes and edges (number of nodes: 4039, number of edges: 88234). Given the non-trivial size of the graph, the author has chosen to employ a bar plot to represent an overview of the nodes with the highest degrees and a *scatter plot* and a *box plot* to illustrate the overall distribution [A.1](#). To determine the true "uniqueness" of specific nodes in terms of "importance" or "famousness" within the network, the author then presents two centrality measurement methods: **Betweenness** and **PageRank** [4]. The precise technical characteristics of these tools are delineated in the respective sections ([2](#), [3](#)).

The author took the opportunity to extend beyond the requirement of the task, mainly through the visual representation [A.1](#) and the code documentation [3] in the interest of gaining in-depth knowledge of the courses' topics and fulfilling personal intellectual curiosity.

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1 Betweenness Centrality [link](#) · [index](#)

This type of centrality falls in the **path-based** category. In fact, as also shown from formula 1, this centrality measure declines the *importance* of a node as the proportion of *shortest* paths between *all* the nodes that have such node as one of their steps.

For a given node v of an undirected graph, its centrality can be measured as:

$$C_B(v) = \sum_{\{s,t\} \subset V \setminus \{v\}} \frac{\sigma_{st}(v)}{\sigma_{st}} \quad (1)$$

Where σ_{st} and $\sigma_{st}(v)$ represent, respectively, the number of shortest paths between nodes s and t , and how many of them (in particular) pass through node v . Such formulation consequently requires a **search algorithm** such that, for each pair of (connected) nodes, computes the shortest path and the list of nodes in it.

1.1 Implementation

We decided to use **Brandes' Algorithm** [4] implementation. In a network composed of n nodes, this method has a time complexity of $O(n \times m)$ [5] as opposed to the Dijkstra's Algorithm which would add up to $O(n^3)$ [4]. The formulation for this case becomes:

$$C_B(v) = \sum_{s \neq v} \delta_s(v) \quad \delta_s(v) = \sum_{w: v \in P_s(w)} \frac{\sigma_{sv}}{\sigma_{sw}} (1 + \delta_s(w)) \quad (2)$$

Where $\delta_s(v)$ is the *dependency* of a source node s on the node of interest v , i.e., the fraction of shortest paths starting at s that pass through it, accumulated recursively from the successors w in each of those paths. The index $w : v \in P(w)$ in the second summation iterates over all nodes w for which v is a predecessor on some shortest path from s to w . Additional details on the programmatic implementation of this algorithm can be found in the source code repository [3].

1.2 Results

The implementation of the algorithm previously mentioned returned the following results:

Rank	1	2	3	4	5	6	7	8	9	10
Node	107	1684	3437	1912	1085	0	698	567	58	428
Score	0.4805	0.3378	0.2361	0.2293	0.1490	0.1463	0.1153	0.0963	0.0844	0.0643

Table 1: Top 10 nodes by Betweenness centrality.

As we can see from the table, the first four nodes (107, 1684, 3437, 1912) have scores that clearly dominate those of all the other nodes, indicating that they concentrate most of the shortest-path "traffic" in the network, therefore acting as critical brokers. It's also worth noting that the scores lie in the range $[0, 1]$ since they are normalized with respect to the maximum attainable betweenness in a graph of this size.

2 PageRank Centrality [link](#) · [index](#)

This type of centrality falls in the **spectral** category. As also shown from formula 3, this centrality measure declines the *importance* of a node as a "discounted" value of their *neighbors* importance. The discount here relates to the degree of such node, as a way of saying that an important node with many connections won't make all equally as important as it, but each will get only a fraction of its importance.

For a given node v its centrality can be measured as:

$$C_{PR}(v) = \alpha \sum_u A_{uv} \frac{C_{PR}(u)}{\deg(u)} + \beta \quad (3)$$

Where α and β represent, respectively, the *damping factor* and the *teleportation factor* as equivalent of how much to consider the *internal* network structure and the *external knowledge* in the calculation.

2.1 Implementation

We decided to use **Power Iteration** implementation. In a network composed of n nodes, this method has a time complexity of $O(n)$ [6] as opposed to the vector formulation (omitted for brevity) which, in order to compute the centrality of all nodes, requires a matrix inversion step of complexity to $O(n^3)$ [7]. The pseudo-code for this case becomes:

1. Initialize $\mathbf{C}^{(0)}$ with initial values (e.g., uniform distribution)
2. For each node, redistribute centrality to its neighbors: $C^{(k+1)}(v) = \alpha \sum_{u \in \mathcal{N}(v)} \frac{C^{(k)}(u)}{\deg(u)} + \beta$
3. Repeat until convergence: $\|\mathbf{C}^{(k+1)} - \mathbf{C}^{(k)}\| < \epsilon$

As opposed to formula 3, here we are also avoiding to compute A_{uv} which, considering the number of nodes and edges would have resulted extremely *sparse*. Additional details on the programmatic implementation of this algorithm can be found in the source code repository [3].

2.2 Results

The implementation of the algorithm previously mentioned returned the following results:

Rank	1	2	3	4	5	6	7	8	9	10
Node	3437	107	1684	0	1912	348	686	3980	414	483
Score	0.0076	0.0069	0.0063	0.0062	0.0038	0.0023	0.0022	0.0022	0.0018	0.0013

Table 2: Top 10 nodes by PageRank centrality.

As we can see from the table, the four most central nodes (3437, 107, 1684, 0) attract a large share of the random walk "importance", reflecting their central role in the network's link structure. It's also worth noting that the scores sum up to 1 since each represents a long-run probability that a random surfer is found at that node.

3 Conclusions [link](#) · [index](#)

The implementation of these two methods in Sections 1.1 and 2.1 offered a comprehensive and hands-on exploration of the underlying logic of these measures.

While the Betweenness method 1 focuses on shortest paths connecting nodes that are distant from each other, PageRank 2 instead emphasizes adjacent nodes, redistributing centrality recursively and incorporating external information. In the context of a *social* network, for instance, PageRank considers the importance of a person within their local community. It is interesting to note that this last centrality measure was also central to Google's first implementation [8].

Despite their fundamental difference, the two methods resulted in **converging outputs** since five nodes are common to both implementations (107, 1684, 3437, 1912, 0). This is a source of extreme interest: even though the two methods pertain to different categories (shortest-path and spectral, respectively), they ended up agreeing on a higher-level notion of "centrality" that goes beyond the specific implementation. The mentioned nodes all are also part of the top 10 nodes with the highest degree, indicating that degree-supported evidence is still retained in the results of the centrality calculations.

Another useful comparison concerns the substantial difference in **computational cost**: the Betweenness method required approximately *4 minutes* of computation on an M1 CPU, whereas the PageRank method required a mere *1 second*. An improvement for the first method could involve considering only a subset of nodes' shortest paths or implementing a more efficient all-to-all search algorithm.

As a safety check, the NetworkX [2] methods (`nx.betweenness_centrality` and `nx.pagerank`) were used to verify the correctness of the hand-implemented algorithms ensuring that the differences remained below a small tolerance threshold.

In summary, the present study indicates that relying on a *unique* centrality measure may be misleading, since different formulations emphasize different *definitions* of "importance". At the same time, these methods tend to agree on certain nodes regardless of their different implementation, leading to a higher-level definition of importance.

A Appendix

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A.1 Visualizations

A.1.1 Network Overview

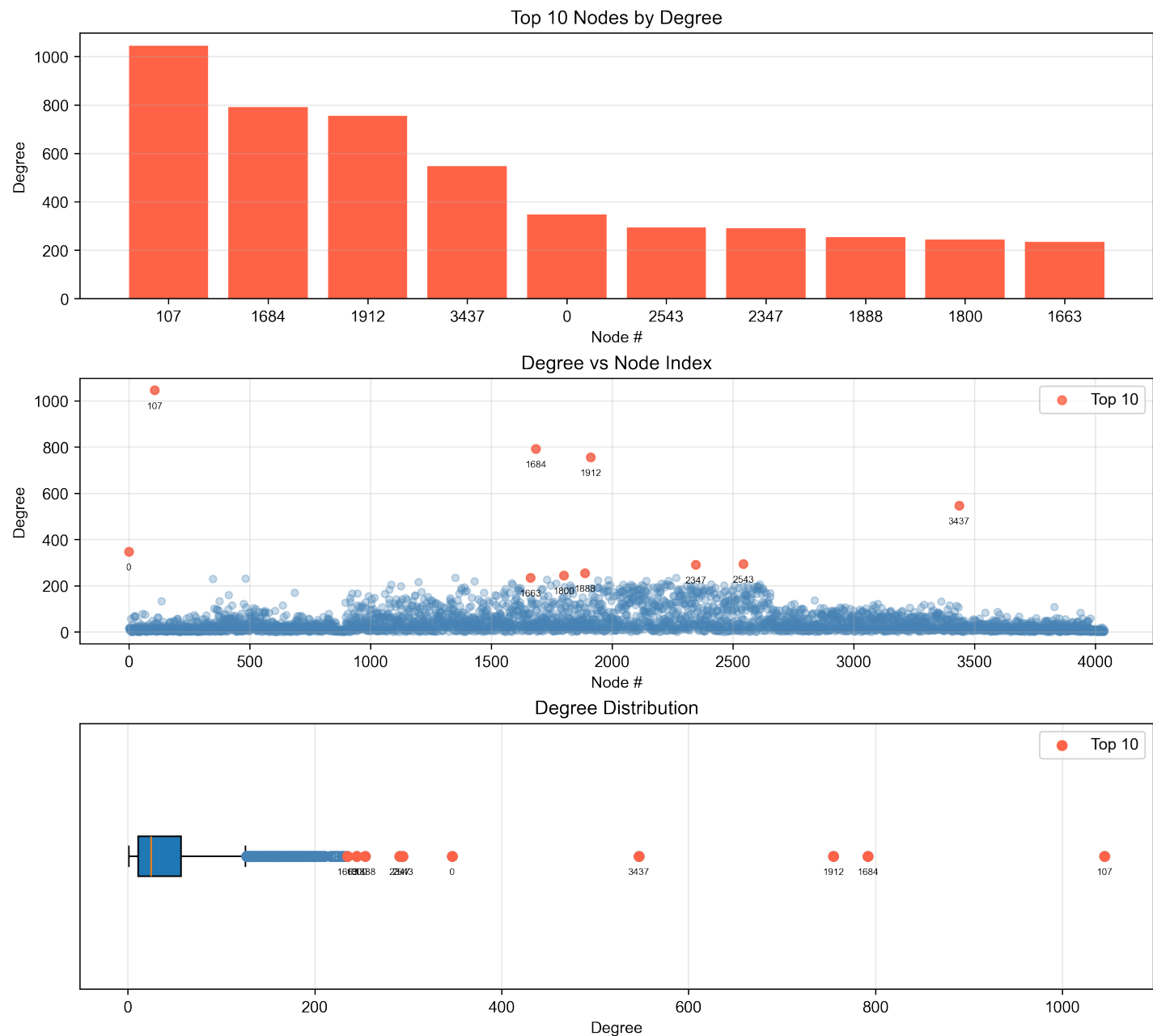


Figure 1: Network overview

A.1.2 Centrality Results

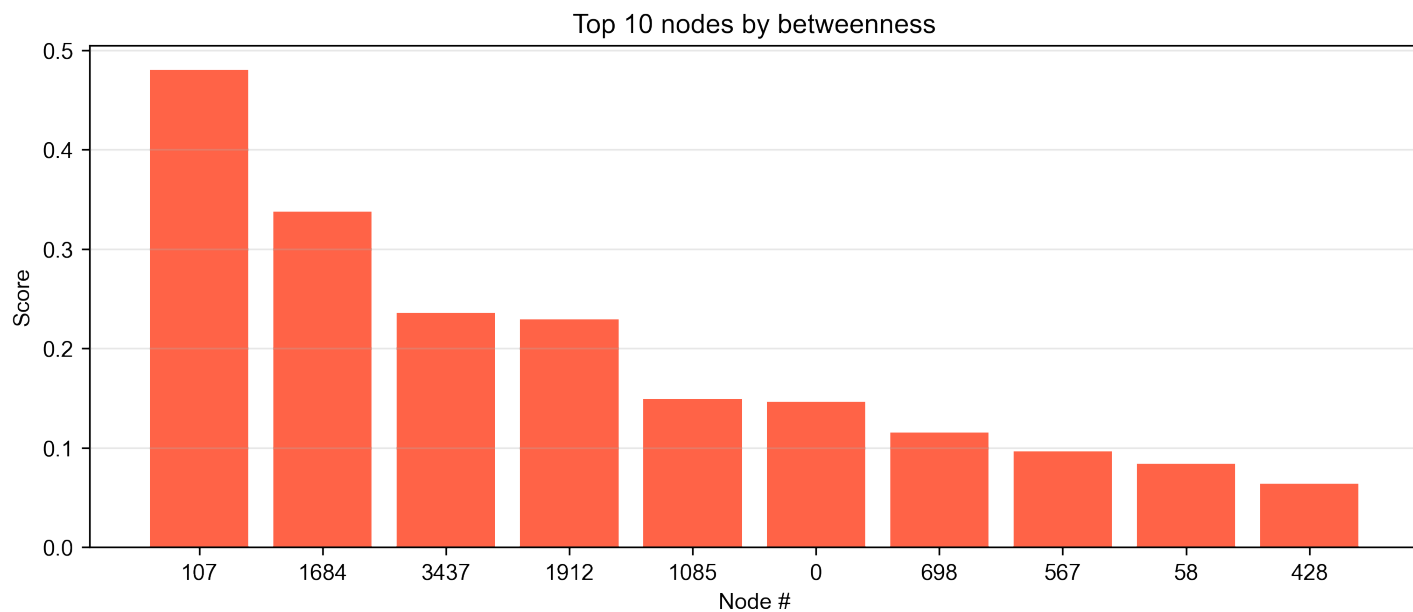


Figure 2: Top 10 nodes by Betweenness centrality

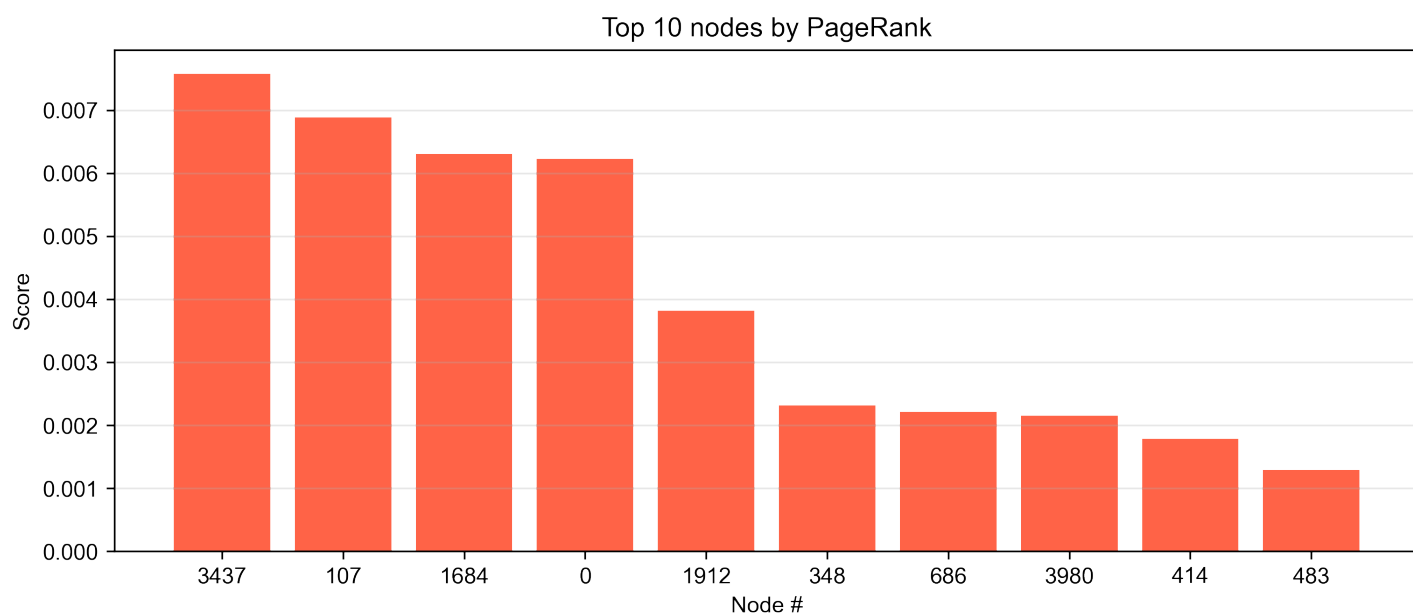


Figure 3: Top 10 nodes by PageRank centrality

A.2 About Generative AI Usage

In the perspective of fair and ethical usage of LLMs, we state that the usage of GenAI has been limited to support tool for writing and debugging the source code and to correct grammatical mistakes in the text. In writing the present document, we felt the commitment to maintain the genuineness of our handwritten text as much as possible.

The following list presents examples GenAI model prompts as part of our workflow:

- **Proofreading:** "Is this paragraph grammatically correct? Is there a more cohesive way of putting it to words? (...)"
- **Clarification:** "Please explain the content of this slide. (...)"
- **Understanding:** "If method A and method B relate to the same category, why are their values differing so much? (...)"
- **Debugging:** "Why is this snippet taking such a long time to complete? Is there any problem? (...)"
- **Best practices:** "Is the way used in this snippet the most community-supported way of solving the problem? Are there any better ways? (...)"

References

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